9th Class • Punjab Board • PECTAA New Syllabus

Chapter 4: Turning Effects of Force

Complete solved exercise — torque, equilibrium, centre of gravity, stability, couples and centripetal force.

Chapter 4 • Complete official exercise • 10 MCQs + 10 short + 5 constructed + 4 comprehensive + 10 numericals

What this chapter is about

• **Torque** τ = rF sin θ = force × perpendicular moment arm. It is measured in N m, not joules in this context.
• **Equilibrium** requires ΣF = 0 and Στ = 0. A couple has ΣF = 0 but produces rotation because its forces have different lines of action.
• **Centre of gravity** is the point where total weight may be considered to act. Stable design uses a low centre of gravity and a broad base.
• **Centripetal force** is the inward resultant needed for circular motion: F_c = mv²/r, always perpendicular to tangential velocity.

Convention: Anticlockwise torque is taken positive and clockwise negative. Resolve angled forces before applying equilibrium. The see-saw exercise's printed "60 N" is dimensionally incorrect; its SI moments are 600 N m on each side.

A. Multiple Choice Questions

1. A particle is simultaneously acted upon by two forces of 4 and 3 newtons. The net force on the particle is:

(a) 1 N (b) between 1 N and 7 N
(c) 5 N (d) 7 N

Answer: (b) between 1 N and 7 N — Without a stated angle, the resultant magnitude can range from |4 − 3| = 1 N to 4 + 3 = 7 N. It equals 5 N only for perpendicular forces.

2. A force F is making an angle of 60° with x-axis. Its y-component is equal to:

(a) F (b) F sin 60°
(c) F cos 60° (d) F tan 60°

Answer: (b) F sin 60° — The component opposite the given angle is F_y = F sin 60°; the x-component is F cos 60°.

3. Moment of force is called:

(a) moment arm (b) couple
(c) couple arm (d) torque

Answer: (d) torque — Moment of force and torque both mean the turning effect of a force about an axis.

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4. If F_1 and F_2 are the forces acting on a body and τ is the torque produced in it, the body will be completely in equilibrium when:

(a) $\Sigma F = 0$ and $\Sigma \tau = 0$
(b) $\Sigma F = 0$ and $\Sigma \tau \neq 0$
(c) $\Sigma F \neq 0$ and $\Sigma \tau = 0$
(d) $\Sigma F \neq 0$ and $\Sigma \tau \neq 0$

Answer: (a) $\Sigma F = 0$ and $\Sigma \tau = 0$ — Complete equilibrium requires both translational and rotational conditions: zero resultant force and zero resultant torque.

5. A shopkeeper sells his articles by a balance having unequal arms of the pans. If he puts the weights in the pan having shorter arm, then the customer:

(a) loses
(b) gains
(c) neither loses nor gains
(d) not certain

Answer: (a) loses — At balance W_article × longer arm = W_standard × shorter arm, so the article's weight is less than the standard weight used; the customer receives less.

6. A man walks on a tight rope. He balances himself by holding a bamboo stick horizontally. It is an application of:

(a) law of conservation of momentum
(b) Newton's second law of motion
(c) principle of moments
(d) Newton's third law of motion

Answer: (c) principle of moments — He adjusts the stick to control turning moments and keep the system's centre of gravity over the rope.

7. In stable equilibrium, the centre of gravity of the body lies:

(a) at the highest position
(b) at the lowest position
(c) at any position
(d) outside the body

Answer: (b) at the lowest position — A small displacement raises the centre of gravity and creates a restoring moment; its stable position corresponds to minimum potential energy.

8. The centre of mass of a body:

(a) lies always inside the body
(b) lies always outside the body
(c) lies always on the surface of the body
(d) may lie within, outside or on the surface

Answer: (d) may lie within, outside or on the surface — It can be inside a solid sphere, on a suitable surface distribution, or outside material as in a ring or boomerang.

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9. A cylinder resting on its circular base is in:

(a) stable equilibrium (b) unstable equilibrium
(c) neutral equilibrium (d) none of these

Answer: (a) stable equilibrium — On its flat circular base, a small tilt raises its centre of gravity and gravity returns it to the base while its vertical projection stays within the support.

10. Centripetal force is given by:

(a) rF (b) rF cos θ
(c) mv^2/r (d) mv/r^2

Answer: (c) mv^2/r — For uniform circular motion, the inward resultant force has magnitude $F_c = mv^2/r = mr\omega^2$.

B. Short Answer Questions

4.1. Define like and unlike parallel forces.

Like parallel forces are parallel and act in the same direction. **Unlike parallel forces** are parallel but act in opposite directions; their magnitudes may be equal, as in a couple, or unequal.

4.2. What are rectangular components of a vector and their values?

Rectangular components are two perpendicular vectors, usually along x and y axes, whose vector sum is the original vector. For A at angle θ to +x: $A_x = A \cos\theta$ and $A_y = A \sin\theta$; $A = \sqrt{(A_x^2 + A_y^2)}$, $\tan\theta = A_y/A_x$.

4.3. What is the line of action of a force?

The line of action is the straight line along which the force acts, extended in both directions. The perpendicular distance from the pivot to this line is the moment arm.

4.4. Define moment of a force. Prove that τ = rF sin θ, where θ is the angle between r and F.

The **moment (torque) of a force** about an axis is its turning effect. It equals force multiplied by the perpendicular distance from the axis to the force's line of action.

Only the perpendicular component $F_\perp = F \sin\theta$ turns the position vector r.

$$\tau = rF_\perp = rF \sin\theta$$

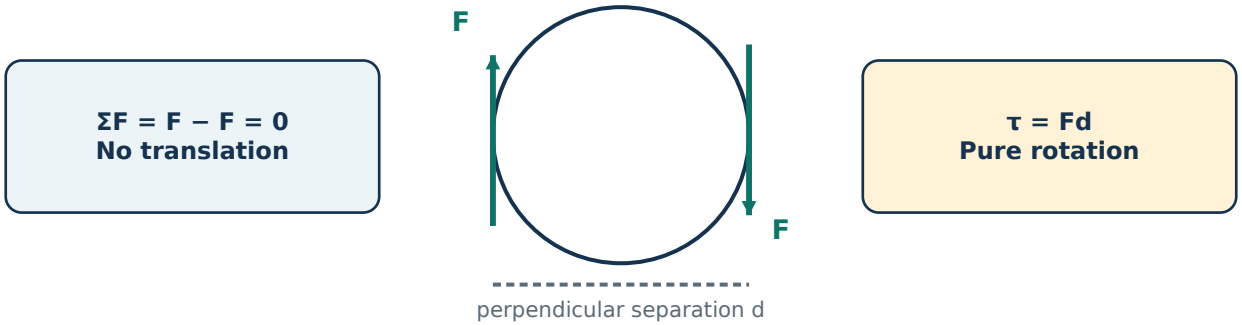
Equivalently, moment arm $d = r \sin\theta$, so $\tau = Fd$

The component F cos θ passes radially through the pivot and produces no moment. Maximum torque occurs at θ = 90° and zero torque at 0° or 180°.

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4.5. With the help of a diagram, show that the resultant force is zero but the resultant torque is not zero.

Figure 1 — Equal and opposite forces separated by distance d form a couple



The two equal, opposite and parallel forces have different lines of action.

Resultant force = F − F = 0

Each force produces torque F(d/2) in the same rotational sense about the centre.

Resultant torque = F(d/2) + F(d/2) = Fd ≠ 0

Steering wheels and screwdriver handles use couples to produce rotation without net translation.

4.6. Identify the state of equilibrium in each case: (a) cone resting on its base (b) sphere on top of a convex surface (c) cylinder lying on its curved side.

(a) **Stable equilibrium:** a small displacement raises the centre of gravity and gravity restores it. (b) **Unstable equilibrium:** a small displacement lowers the centre of gravity and the sphere moves away. (c) **Neutral equilibrium:** rolling horizontally leaves the centre of gravity at the same height.

4.7. Give an example of the body which is moving yet in equilibrium.

A car moving in a straight line at constant velocity on a level road is in dynamic translational equilibrium: driving and resisting forces balance, so ΣF = 0 and acceleration is zero. A non-accelerating wheel system must also satisfy Στ = 0 for complete equilibrium.

4.8. Define centre of mass and centre of gravity of a body.

Centre of mass is the point at which the whole mass may be treated as concentrated for translational motion. **Centre of gravity** is the point through which the resultant weight acts. In a uniform gravitational field, the two points coincide; the centre of mass need not lie within material.

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4.9. What are two basic principles of stability in physics which are applied in designing balancing toys and racing cars?

Keep the **centre of gravity as low as possible** and make the **base of support as wide as possible**. A low centre of gravity requires more lifting before toppling, while a wide base lets the vertical line through the centre of gravity remain inside support during greater tilts.

4.10. How can you prove that the centripetal force always acts perpendicular to velocity?

In uniform circular motion speed v is constant, so $v^2 = v \cdot v$ is constant. Differentiating gives $2v \cdot a = 0$, hence $v \cdot a = 0$. Therefore acceleration is perpendicular to velocity; since $F_c = ma$, centripetal force is also perpendicular to velocity and directed toward the centre.

C. Constructed Response Questions

4.1. A car travels at the same speed around two curves with different radii. For which radius does the car experience more centripetal force? Prove your answer.

$$F_c = mv^2/r$$

$$\text{At fixed } m \text{ and } v: F_c \propto 1/r$$

The **smaller-radius curve** requires the larger centripetal force. For example, halving r doubles F_c. If tyre–road friction cannot provide this inward force, the car cannot follow the intended curve.

4.2. A ripe mango does not normally fall from the tree. But when the branch of the tree is shaken, the mango falls down easily. Can you tell the reason?

When the branch moves suddenly, the mango tends to retain its previous state because of inertia. The stalk may be unable to supply the force needed to accelerate the mango with the branch, so the weakened ripe stalk breaks. Gravity then pulls the detached mango downward.

4.3. Discuss stability and centre of gravity in relation to objects toppling over. Provide an example and explain how altering the base of support influences stability.

An object remains stable while the vertical line through its centre of gravity falls inside its base of support. As it tilts, the line moves toward an edge; once it passes outside, weight produces a toppling moment rather than a restoring moment.

A loaded high vehicle has a higher centre of gravity and topples at a smaller tilt than a low racing car. Widening the base moves the support edge outward, increases the

angle through which the centre of gravity must be raised and therefore improves stability.

• Lower heavy components to reduce centre-of-gravity height.

• Spread wheels, legs or supports to widen the base.

• Keep loads centred so the weight line remains safely inside the base.

4.4. Why can an accelerated body not be considered in equilibrium?

Translational acceleration means $\Sigma F = ma \neq 0$, violating the first condition $\Sigma F = 0$. Angular acceleration means $\Sigma \tau \neq 0$, violating the second condition. Equilibrium means no change in translational or rotational velocity, even if the body is moving uniformly.

4.5. Two boxes of the same weight but different heights are lying on the floor of a truck. If the truck makes a sudden stop, which box is more likely to tumble over? Why?

The **taller box** is more likely to tumble because its centre of gravity is higher. During braking, its upper part tends to continue forward; the weight line reaches and passes the front support edge at a smaller tilt. The higher centre also gives inertia a larger moment arm about that edge.

D. Comprehensive Questions

4.1. Explain the principle of moments with an example.

Principle: For a body in rotational equilibrium, the algebraic sum of all moments about any point is zero. Thus total clockwise moment equals total anticlockwise moment about the same point.

$$\Sigma \tau = 0 \rightarrow \Sigma \tau_{clockwise} = \Sigma \tau_{anticlockwise}$$

For a lever: $F_1 d_1 = F_2 d_2$

Example: a 300 N child sits 2 m left of a see-saw pivot and a 400 N child sits 1.5 m right. Anticlockwise moment = 300 × 2 = 600 N m; clockwise moment = 400 × 1.5 = 600 N m. They balance.

Distances must be perpendicular moment arms, not merely distances along an inclined object.

4.2. Describe how you could determine the centre of gravity of an irregular-shaped lamina experimentally.

1. Make three small holes A, B and C near separated points on the edge of the lamina.

2. Suspend the lamina freely from a pin through A and hang a plumb line from the same support.

3. When motion stops, draw the vertical line along the plumb string on the lamina. The centre of gravity lies somewhere on this line.

4. Repeat using B and C to obtain two more vertical lines.

5. Their common intersection is the centre of gravity. Balancing the lamina horizontally at this point provides a check.

At each suspension, the centre of gravity settles vertically below the support so that weight produces no net turning moment.

4.3. State and explain two conditions of equilibrium.

1. First condition—translational equilibrium: The vector sum of external forces is zero, $\Sigma F = 0$. In component form $\Sigma F_x = 0$ and $\Sigma F_y = 0$, so linear acceleration is zero.

2. Second condition—rotational equilibrium: The algebraic sum of external torques about any axis is zero, $\Sigma \tau = 0$. Therefore clockwise and anticlockwise moments balance and angular acceleration is zero.

A body satisfying both can be at rest (**static equilibrium**) or move with constant translational and rotational velocity (**dynamic equilibrium**). Zero force alone is insufficient: a couple can still rotate the body.

4.4. How can the stability of an object be improved? Give a few examples to support your answer.

• **Lower the centre of gravity:** place heavy parts low in racing cars, ships and balancing toys; store heavy boxes on lower shelves.

• **Increase base width:** use widely spaced wheels, spread the legs of a tripod or crane, and give laboratory stands broad heavy bases.

• **Keep weight over support:** centre a load and avoid leaning beyond the edge; passengers should not crowd one side of a small boat.

• **Avoid top-heavy loading:** a loaded high truck takes bends slowly because its weight line can move outside the wheelbase during a tilt.

Greater stability means a larger displacement and more work against gravity are needed before the centre-of-gravity vertical passes beyond the support base.

E. Numerical Problems

4.1 A force of 200 N is acting on a cart at an angle of 30° with the horizontal direction. Find the x and y-components of the force.

Given: F = 200 N; θ = 30° above horizontal.

$$F_x = F \cos \theta = 200 \cos 30° = 200 \times 0.866 = 173.2 \text{ N}$$

$$F_y = F \sin \theta = 200 \sin 30° = 200 \times 0.5 = 100 \text{ N}$$

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Answer: $F_x = 173.2$ N; $F_y = 100$ N.

4.2 A force of 300 N is applied perpendicularly at the knob of a door. The knob is 1.2 m away from the hinge. What torque is applied? Is it positive or negative?

Given: F = 300 N; r = 1.2 m; θ = 90°; the illustrated opening is anticlockwise.

$\tau = rF \sin\theta = 1.2 \times 300 \times 1 = 360$ N m

Anticlockwise torque is positive by the chosen convention.

Answer: +360 N m (positive).

4.3 Two weights hang from a metre rule: unknown w is 40 cm left of its centre of gravity and 4 N is 30 cm right. If balanced at its centre of gravity, find w.

Given: Pivot at C.G.; $d_w = 40$ cm; $d_4 = 30$ cm; rule's own weight has zero moment arm.

Balancing moments: $w \times 40 = 4 \times 30$

$w = 120/40 = 3$ N

Answer: w = 3 N.

4.4 A see-saw has child A of mass 30 kg sitting 2 m from the pivot and child B of mass 40 kg sitting 1.5 m on the other side. Calculate the total moment on each side and determine whether the see-saw is in equilibrium.

Given: Take $g = 10$ m s^{-2}: $W_A = 300$ N and $W_B = 400$ N.

Moment due to A = $W_A d_A = 300 \times 2 = 600$ N m

Moment due to B = $W_B d_B = 400 \times 1.5 = 600$ N m

Net moment = $600 - 600 = 0$

Answer: 600 N m on each side; net torque = 0, so rotational equilibrium. (If masses are read as kgf, each is 60 kgf m; the printed "60 N" is not an SI moment.)

4.5 A crowbar lifts a box. A downward force of 250 N is applied 30 cm from the fulcrum, while the load acts 5 cm on the other side. How much weight does the other end bear? The crowbar has negligible weight.

Given: F = 250 N; effort arm = 30 cm; load arm = 5 cm.

$$W \times 5 = 250 \times 30$$

$$W = 7500/5 = 1500 \text{ N}$$

Answer: Load/weight = 1500 N.

4.6 A 30 cm long spanner is used to open the nut of a car. If the torque required is 150 N m and the applied force is perpendicular, how much force F should be applied?

Given: r = 30 cm = 0.30 m; τ = 150 N m; $\theta = 90°$.

$$\tau = Fr \rightarrow F = \tau/r$$

$$F = 150/0.30 = 500 \text{ N}$$

Answer: 500 N perpendicular to the spanner.

4.7 A 5 N ball hanging from a rope is pulled right by horizontal force F. The rope makes 60° with the ceiling. Determine F and tension T.

Given: Weight W = 5 N; rope angle to horizontal = 60°.

Vertical equilibrium: $T \sin 60° = 5 \rightarrow T = 5/0.866 = 5.77$ N

Horizontal equilibrium: $F = T \cos 60° = 5.77 \times 0.5 = 2.89$ N

Answer: F ≈ 2.9 N; T ≈ 5.8 N.

4.8 A signboard of weight 200 N is suspended by two vertical steel wires, each 2 m from the centre. Find the tension in each string.

Given: W = 200 N; symmetric tensions T_1 and T_2; equal moment arms = 2 m.

Vertical equilibrium: $T_1 + T_2 = 200$

Moments about centre: $2T_1 = 2T_2 \rightarrow T_1 = T_2$

$2T = 200 \rightarrow T = 100$ N

Answer: $T_1 = T_2 = 100$ N.

4.9 One girl of mass 30 kg sits 1.6 m from the axis of a see-saw. Another girl of mass 40 kg sits on the other side. How far from the axis must she sit for equilibrium?

Given: m_1 = 30 kg; d_1 = 1.6 m; m_2 = 40 kg. g cancels.

$$m_1 g d_1 = m_2 g d_2$$

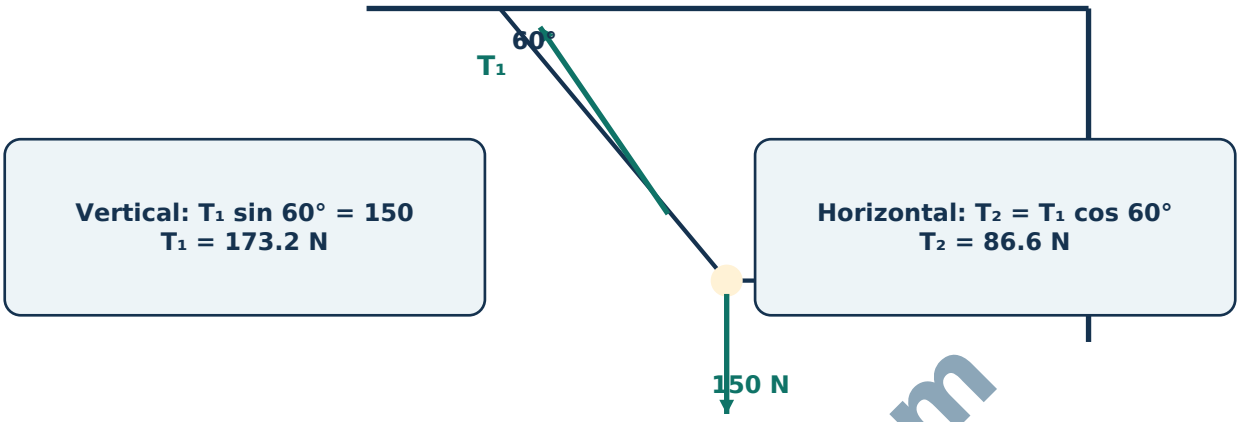
$$d_2 = (30 \times 1.6)/40 = 48/40 = 1.2 \text{ m}$$

Answer: 1.2 m from the axis.

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4.10. Find the tension in each string if the block weighs 150 N. One string is horizontal to a wall; the other rises left and makes 60° with the ceiling.

Figure 2 — Free-body resolution at the knot



Resolve the inclined tension; the horizontal string has no vertical component.

$$T_1 \sin 60° = 150 \rightarrow T_1 = 150/0.866 = 173.2 \text{ N}$$

$$T_2 = T_1 \cos 60° = 173.2 \times 0.5 = 86.6 \text{ N}$$

Inclined string T_1 = 173.2 N; horizontal string T_2 = 86.6 N.

F. More Important Questions (from the text)

1. Why is a door handle fixed far from the hinges?

Torque $\tau = Fr \sin \theta$ is proportional to perpendicular distance r from the hinge. With the handle far from the hinge, the same force produces a larger turning moment, or a required moment can be produced with a smaller force.

The force is most effective perpendicular to the door. Pushing along a line through the hinges gives zero moment arm and therefore no turning effect.

2. Compare stable, unstable and neutral equilibrium.

State	Effect of small displacement	Example
Stable	Centre of gravity rises; restoring moment returns body.	Cone on its base; upright cylinder on flat end.
Unstable	Centre of gravity falls; moment moves body farther away.	Sphere on a convex dome; cone on its tip.
Neutral	Centre of gravity stays at same height; no restoring or toppling moment.	Cylinder or sphere rolling on a level plane.

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Points to Remember

Torque: $\tau = rF \sin\theta = Fd$; SI unit N m.

Moment arm is the perpendicular distance from pivot to force's line of action.

Equilibrium I: $\Sigma F = 0$; in components $\Sigma F_x = \Sigma F_y = 0$.

Equilibrium II: $\Sigma\tau = 0$; clockwise and anticlockwise moments balance.

Couple: equal opposite parallel forces, $\Sigma F = 0$ but $\tau = Fd \neq 0$.

Components: $F_x = F\cos\theta$ and $F_y = F\sin\theta$ when θ is measured from +x.

Centre of gravity is the resultant weight's point of action; it need not lie inside material.

Stability: low centre of gravity and wide support base; toppling begins when weight line passes outside base.

Centripetal force: $F_c = mv^2/r$ inward and perpendicular to velocity.

See-saw: $m_1d_1 = m_2d_2$ because g cancels when distances are perpendicular arms.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	b	2	b	3	d	4	a	5	a
6	c	7	b	8	d	9	a	10	c

How to attempt this chapter in the paper

- Convert centimetres to metres before expressing torque in N m; ratios may use centimetres if both arms use the same unit.
- Take moments about a pivot that removes an unknown force when convenient, but use the same pivot for every moment.
- Resolve an inclined tension into horizontal and vertical components before applying $\Sigma F = 0$.
- Do not write moment in newtons: force × distance gives N m.
- For toppling, compare the vertical line through the centre of gravity with the base edge, not with the geometric centre.
- For circular motion, smaller radius at the same speed requires larger inward force.